

Diabetes en het oog

Oogarts Guy Missotten

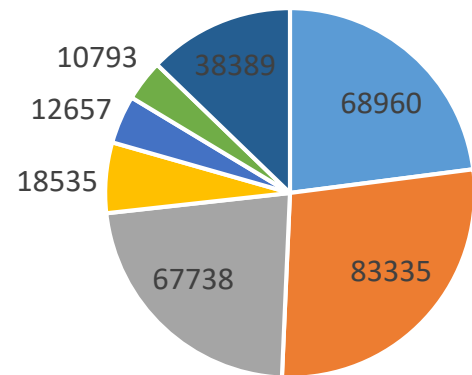
Diabetesvereniging

18 oktober 2017 Virga Jesse Ziekenhuis

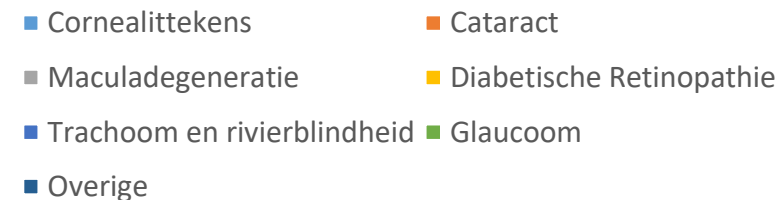
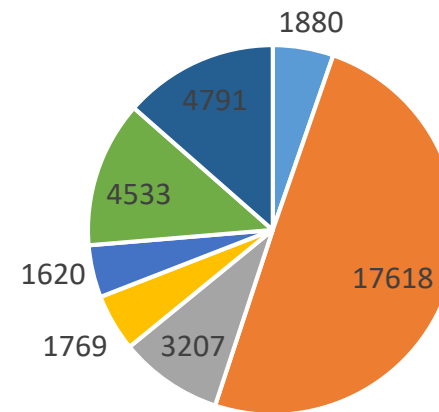


Oorzaken van blindheid en slechtziendheid

- 12 000 blinden in België, en 200 000 mensen zijn slechtziend

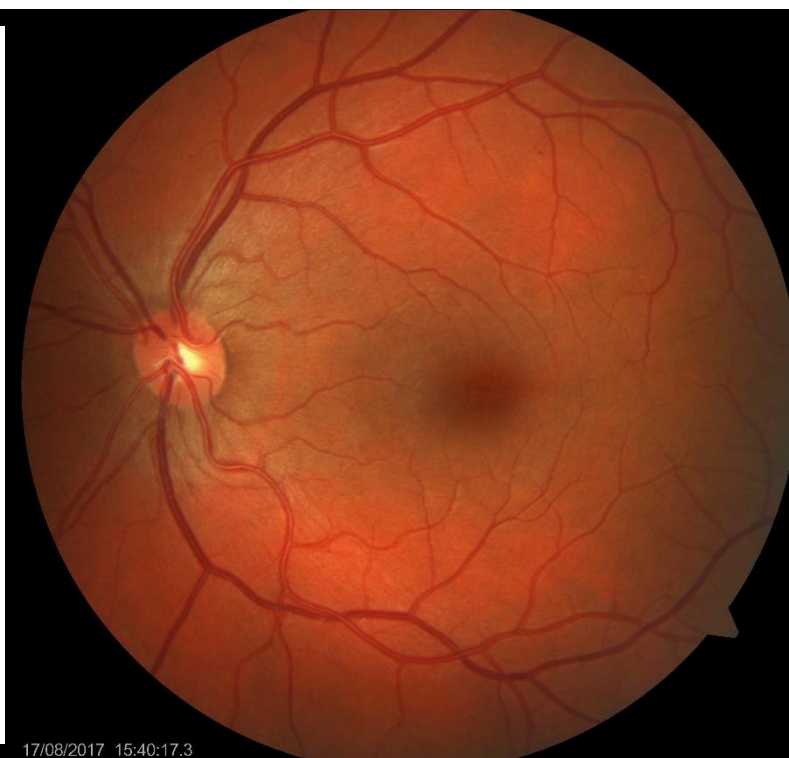
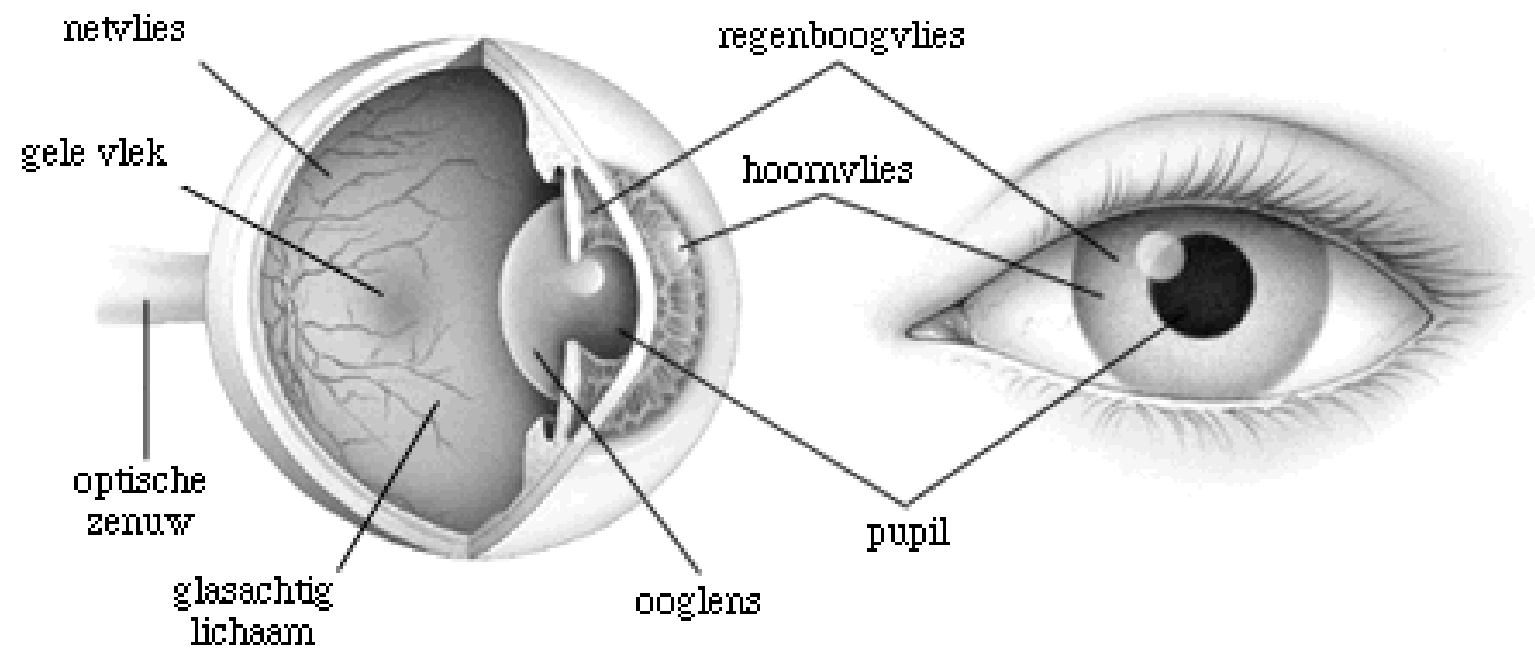


Slechtziendheid in België

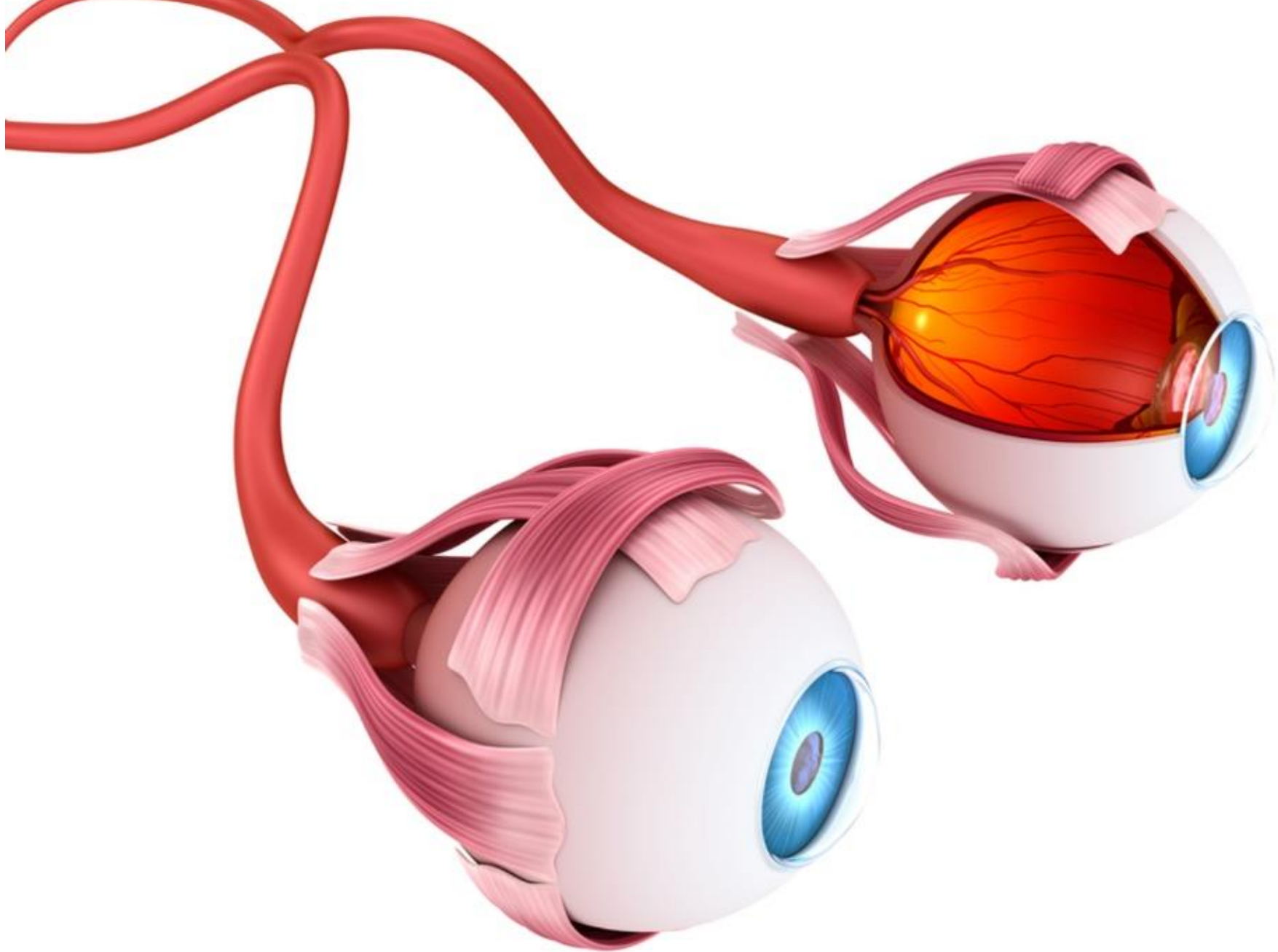


Slechtziendheid in de wereld

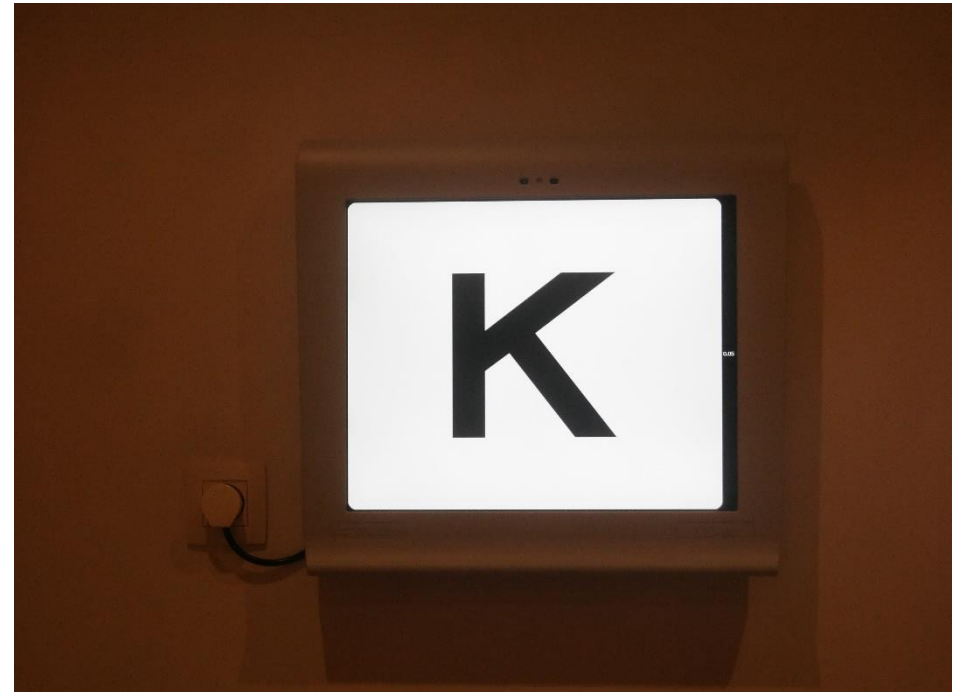
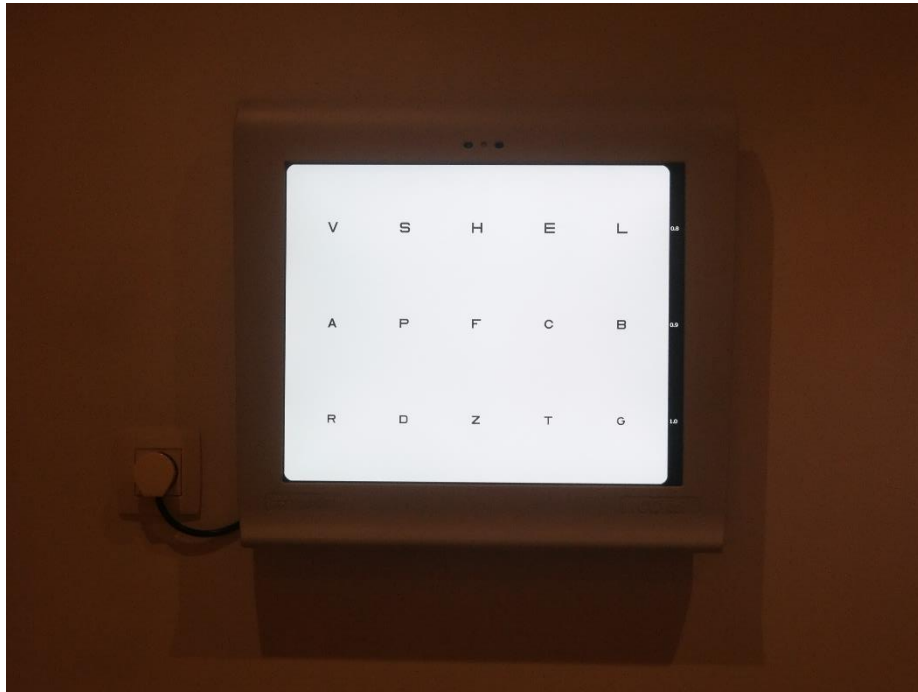
- Diabetische controle-raadpleging
- Mechanisme van diabetische retinopathie en maculopathie
- Enkele concrete patiënten
- Oogkliniek
- Therapie
- Conclusie



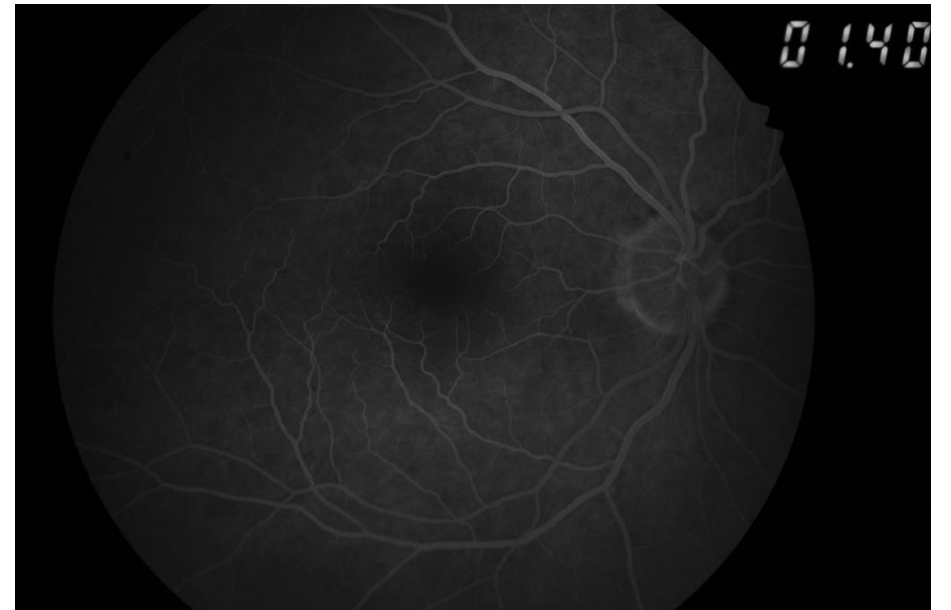
17/08/2017 15:40:17.3



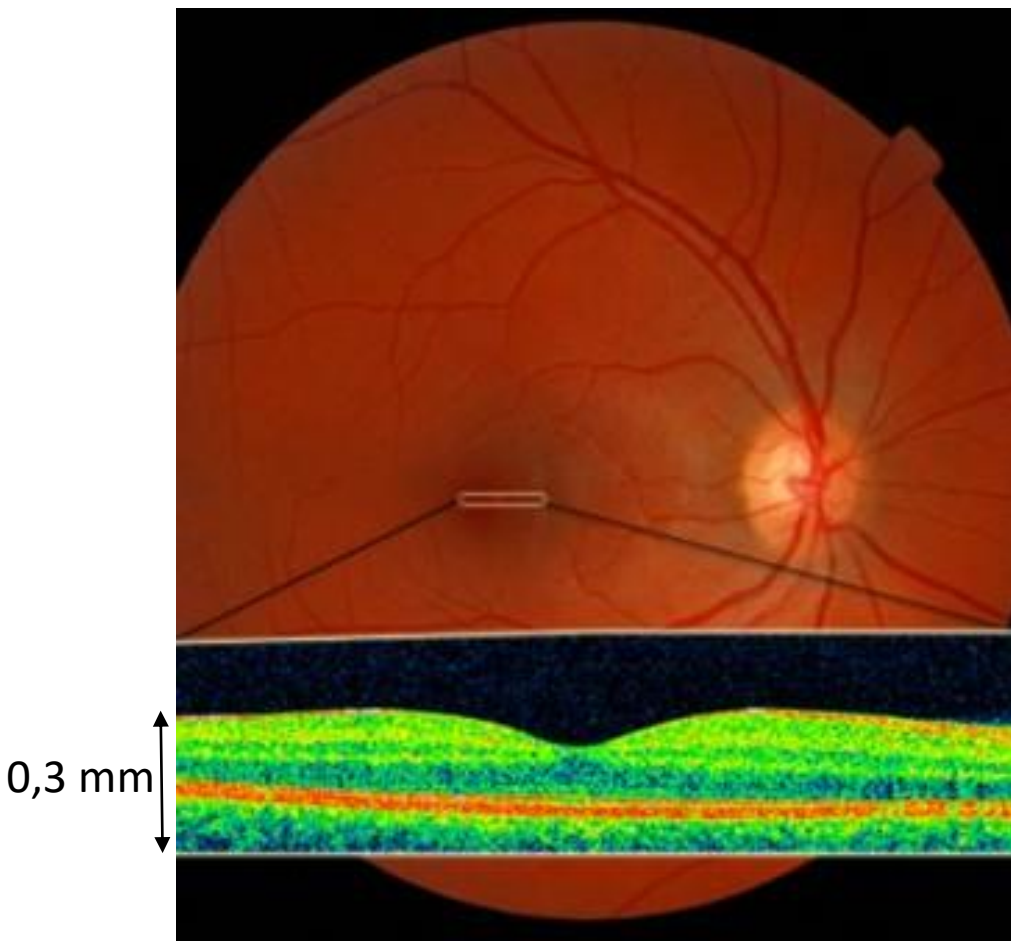
Visusmeting



Fluoangiografie



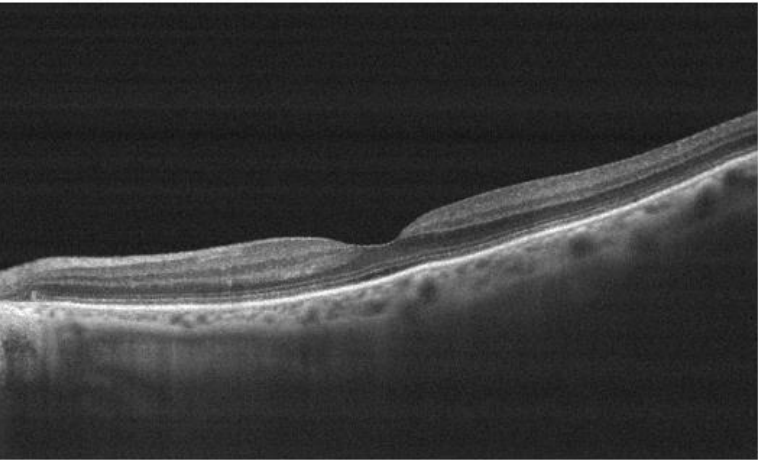
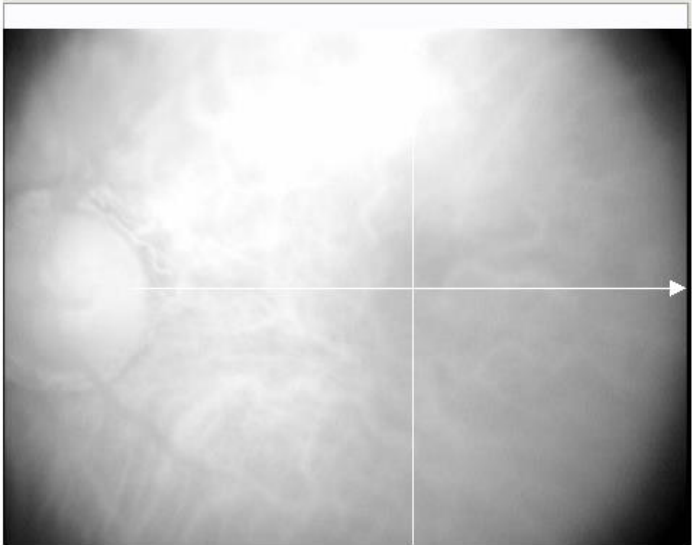
OCT-scan = maculascan



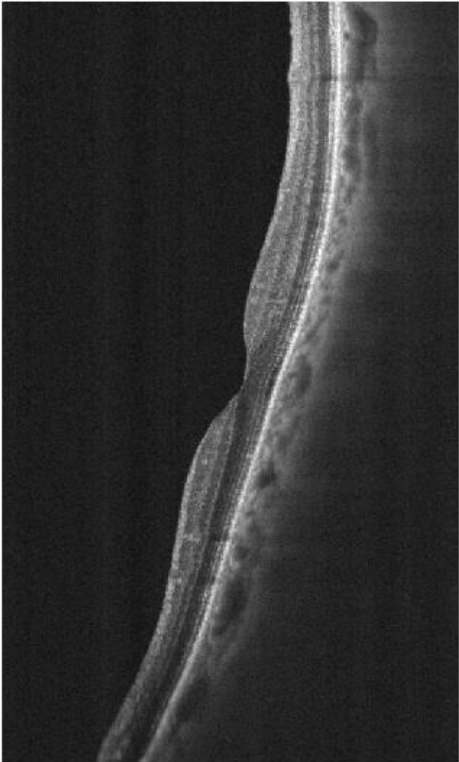
05

Cross Line

SSI= 58.2



8.00mm Scan Length

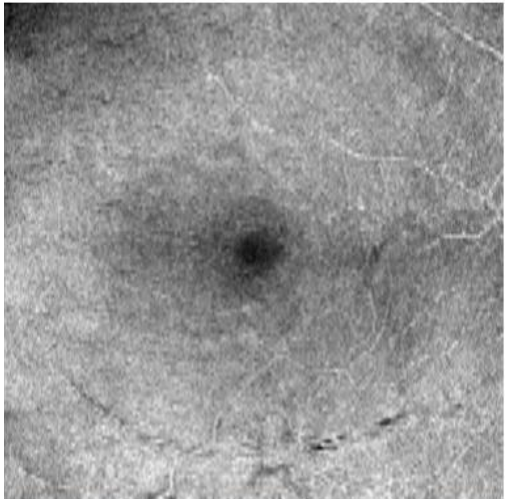


250µm

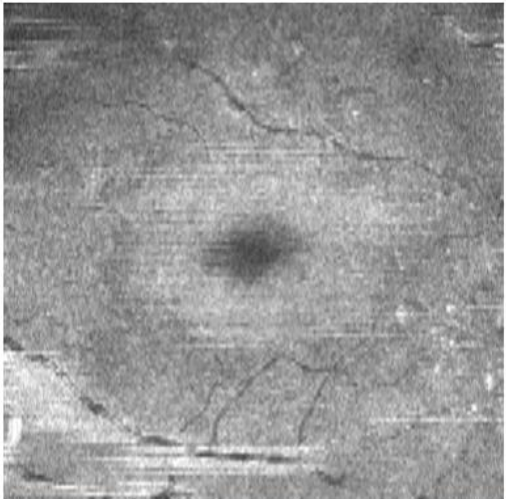
of Averages: 19, 19

Average

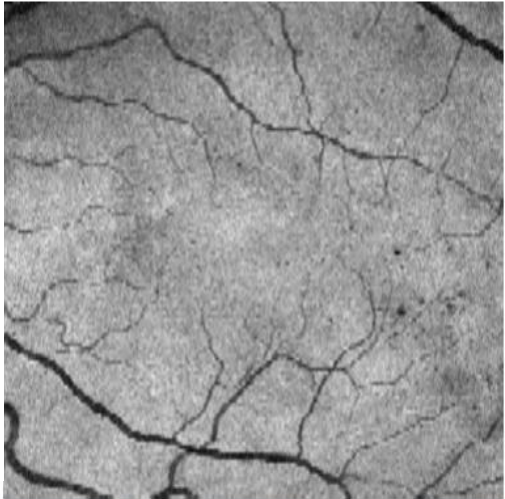
No Average



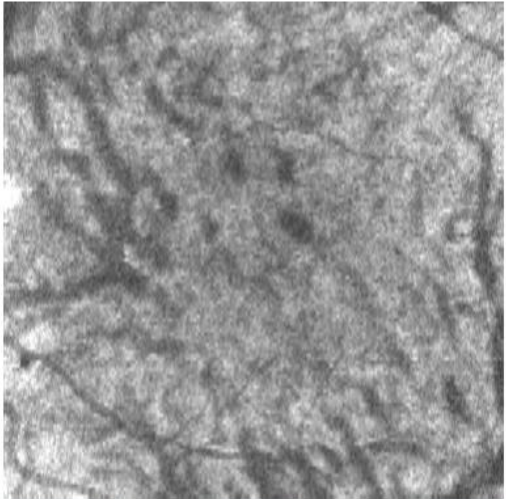
ILM Offset: 0 Thickness: 59



IPL Offset: 0 Thickness: 90



RPE Offset: -59 Thickness: 90



RPE Ref Offset: 90 Thickness: 121

Exit

Lumbrroso

Reset Layers

Set Defaults

Show 3D



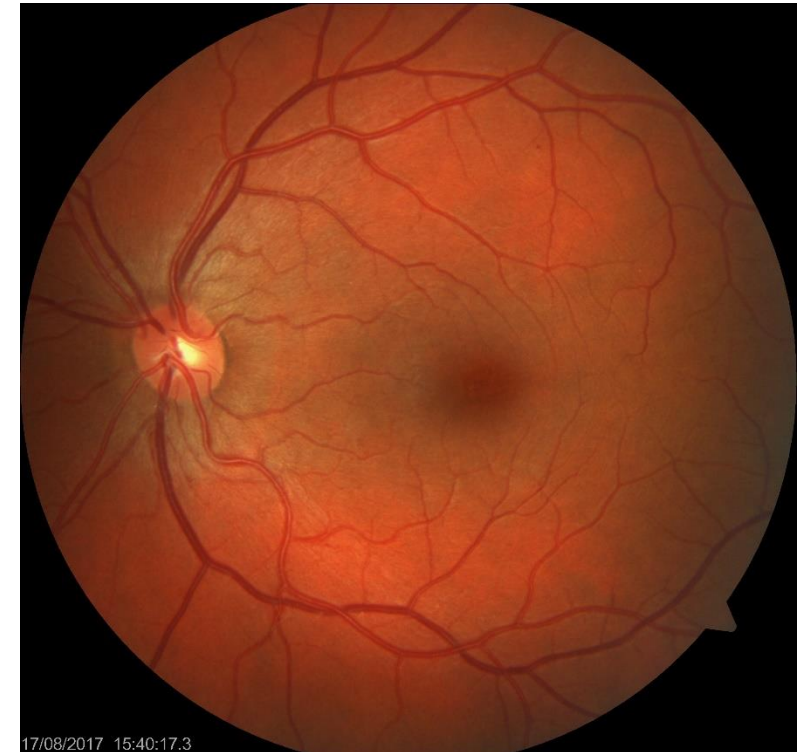
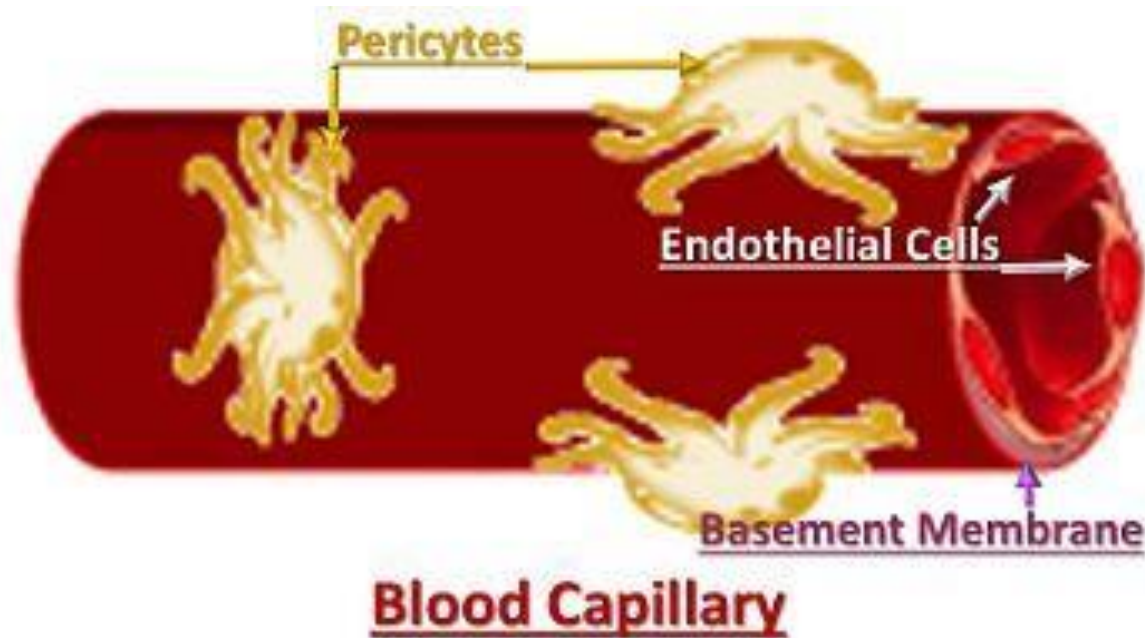
Diabetische retinopathie en maculopathie

- Geen diabetische retinopathie
- Pre-proliferatieve diabetische retinopathie
 - Mild
 - Matig
 - ernstig
- Proliferatieve diabetische retinopathie

+/- maculopathie

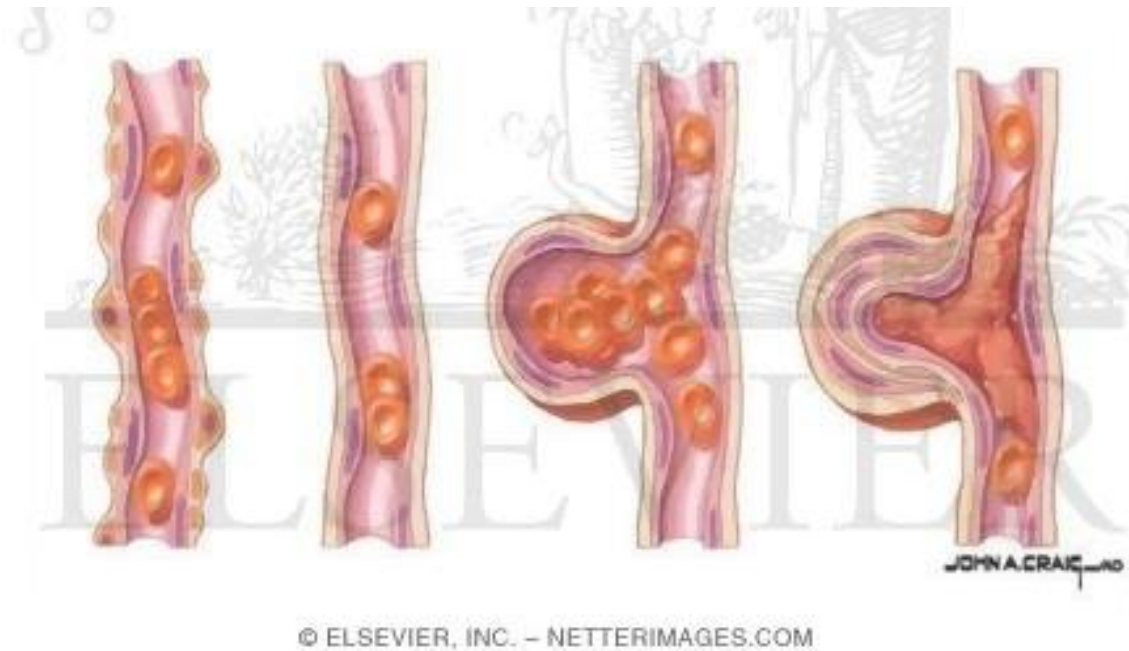
Geen diabetische retinopathie

Door hyperglycemie wordt de wand van kleine bloedvaten aangetast:
verlies van retinale pericyten.

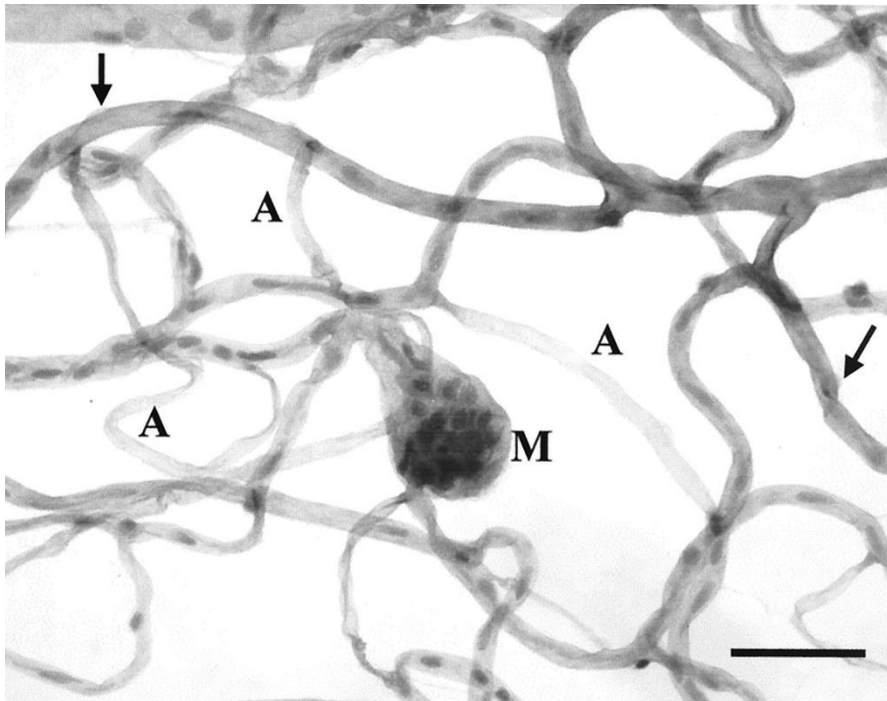


Pre-proliferatieve diabetische retinopathie

Micro-aneurysma: schade door verweken van de vaatwanden

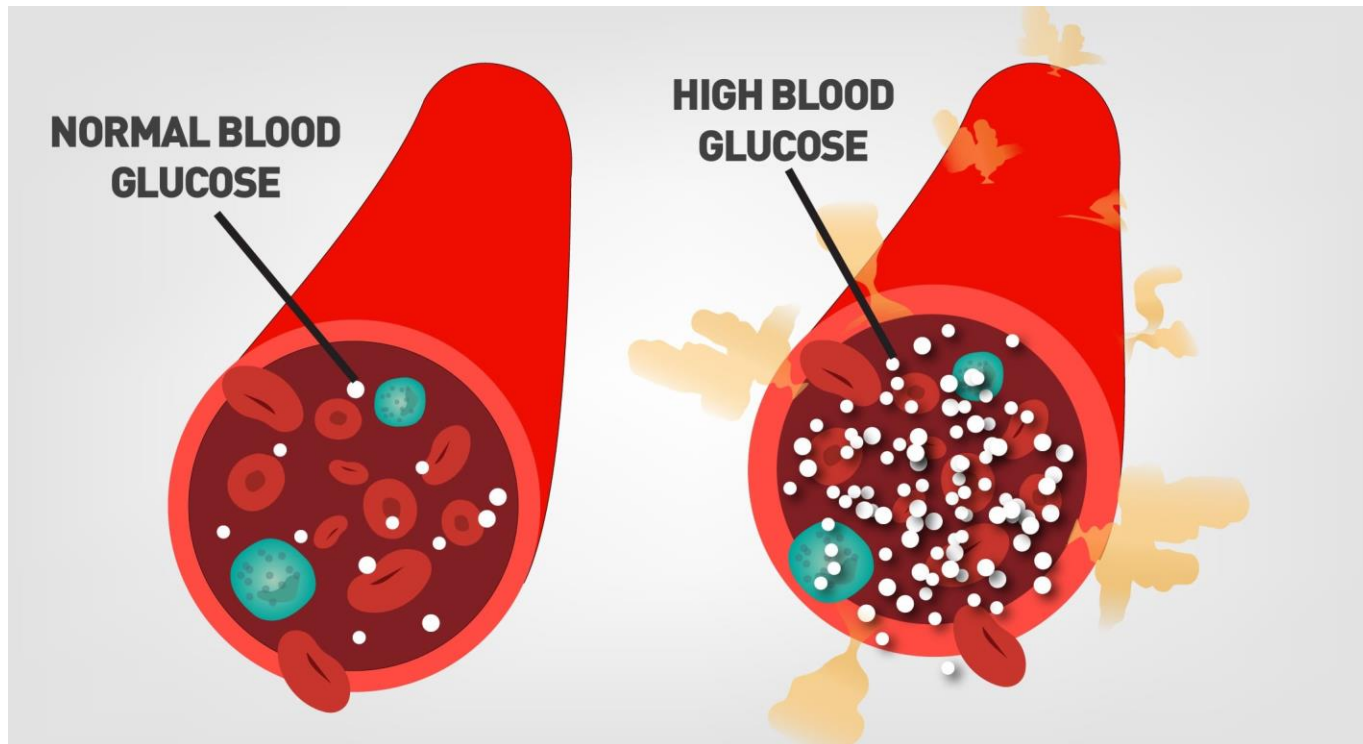


Pre-proliferatieve diabetische retinopathie



Pre-proliferatieve diabetische retinopathie

Toename van de bloedvat permeabiliteit (lekkage):
eiwitten lekken en vetopstapelingen = harde exsudaten



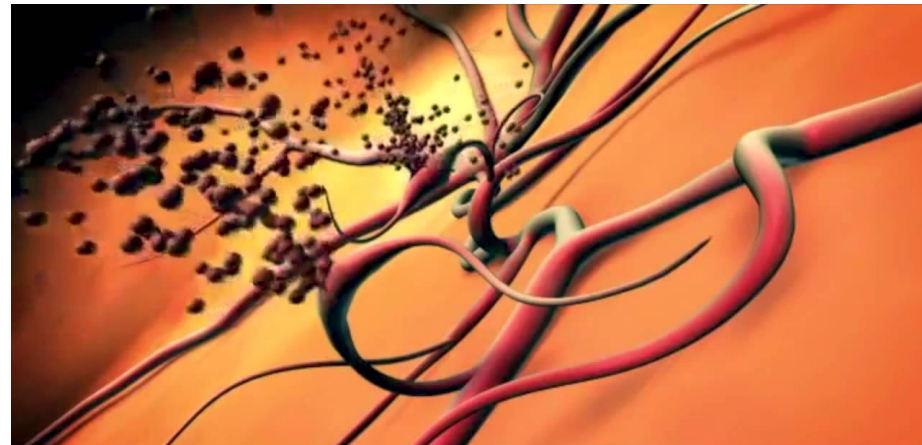
Pre-proliferatieve diabetische retinopathie

Zenuwlaag infarctjes: zachte wollige exsudaten (nerve fiber layer infarcts of cotton wool exsudates)



Pre-proliferatieve diabetische retinopathie

Lekken van bloedcellen: kleine bloedinkjes ontstaan



Pre-proliferatieve diabetische retinopathie

Klasse	Afwijkingen in de retina (netvlies)
Milde	Enkel microaneurysmata
Matige	
Ernstige	Nog geen proliferatieve diabetische retinopathie en >20 intraretinale bloedinkjes in het oog Uitgezette venen in 2 quadranten Intraretinale abnormale vaten (IRMA) in meer dan 1 quadrant

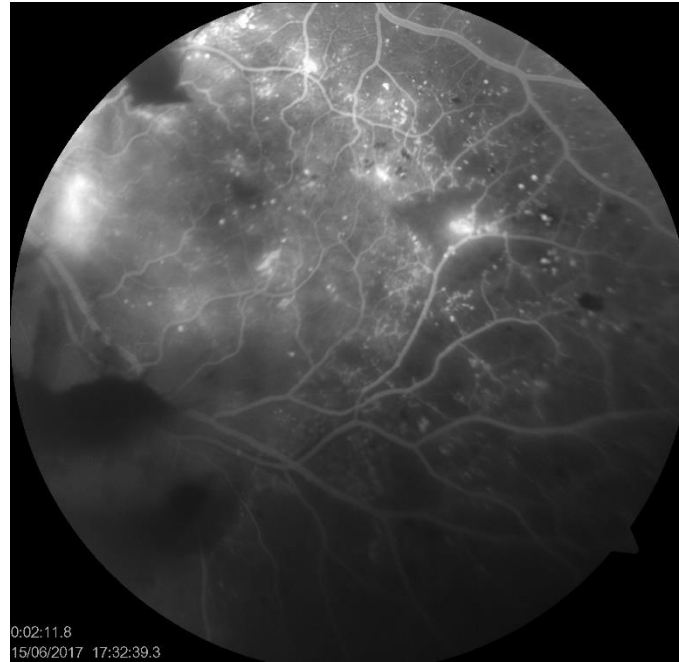
Proliferatieve diabetische retinopathie

Zuurstoftekort leidt tot bloedvat-proliferatie (groei):



Proliferatieve diabetische retinopathie

Bloedingen in het glasvocht

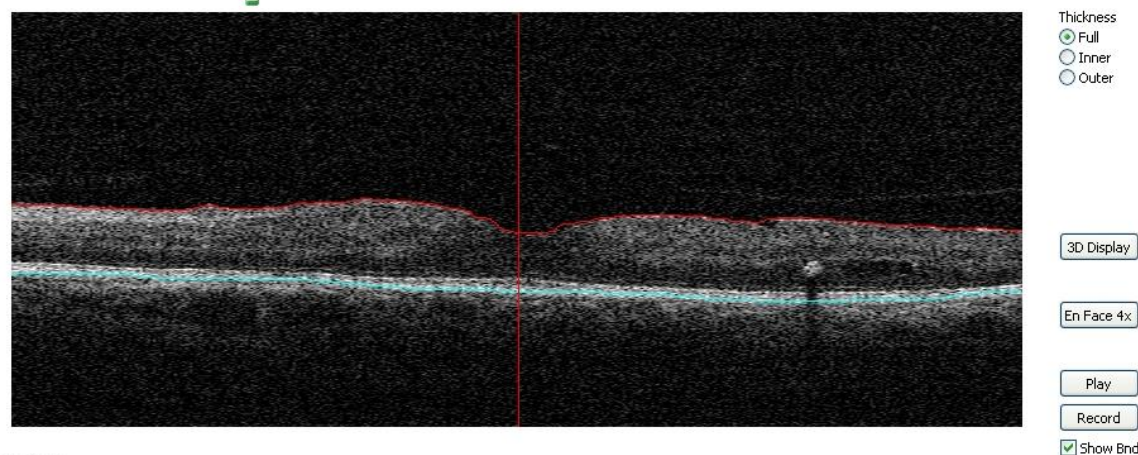
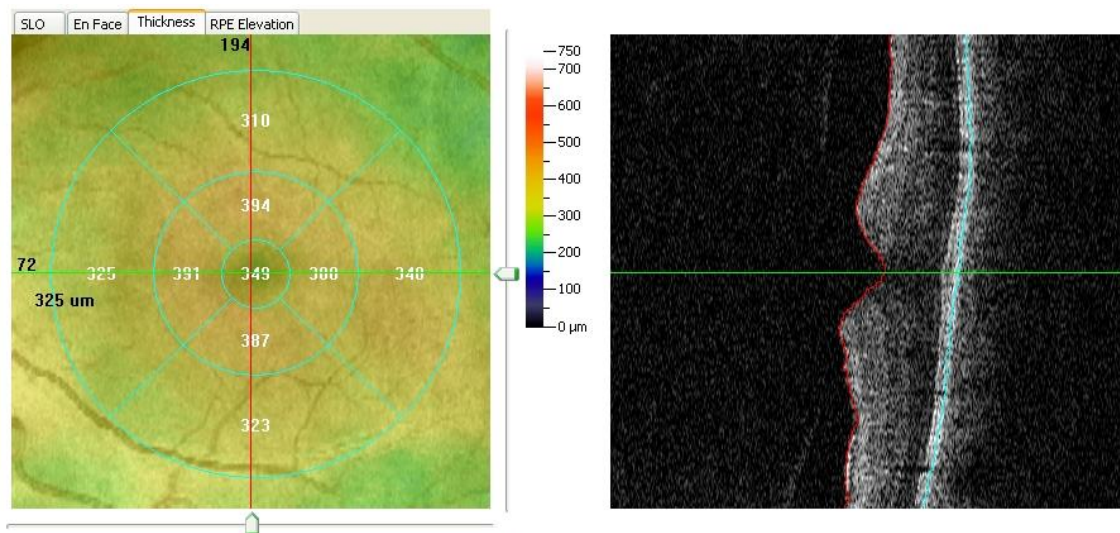
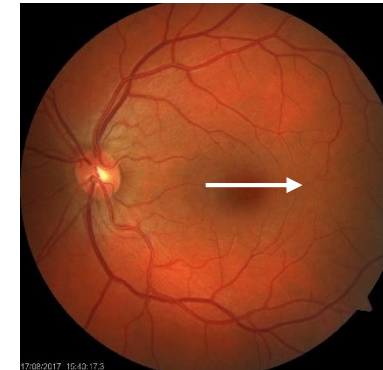


Proliferatieve diabetische retinopathie

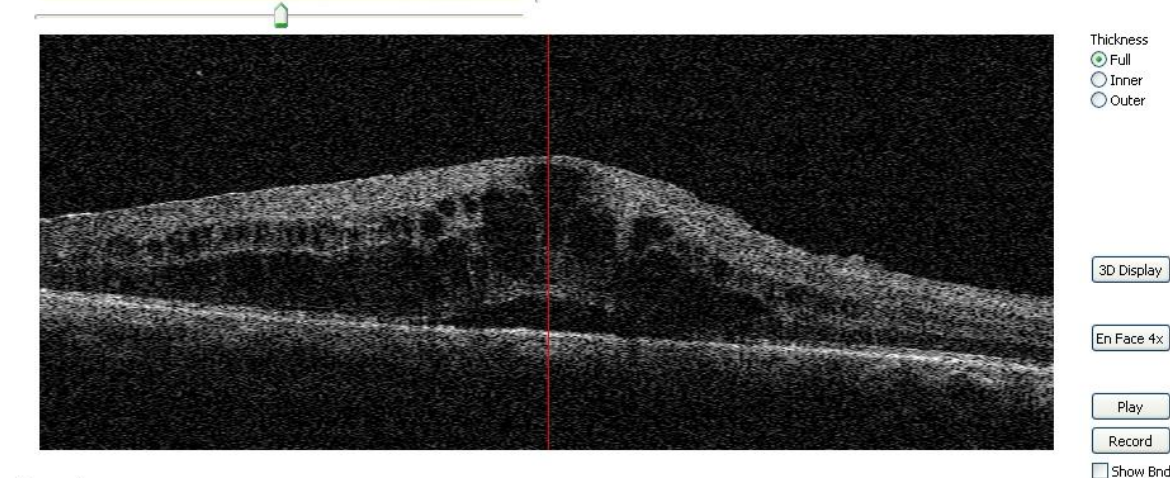
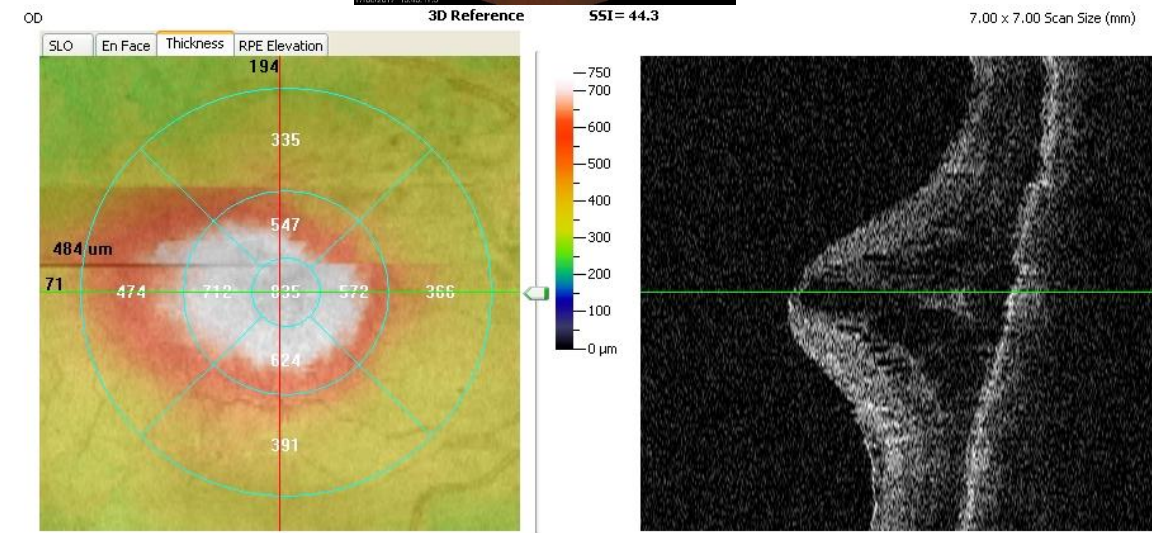
Retinalosclating



Diabetische maculopathie

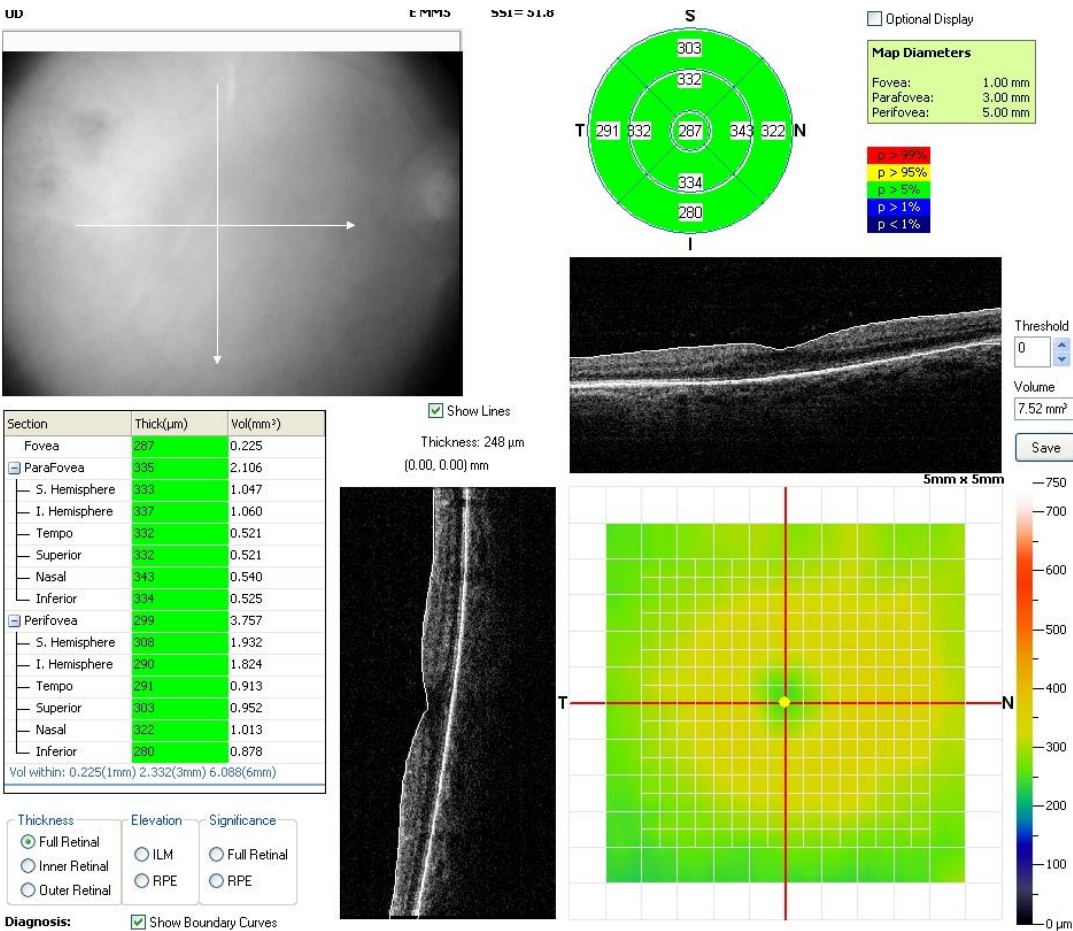
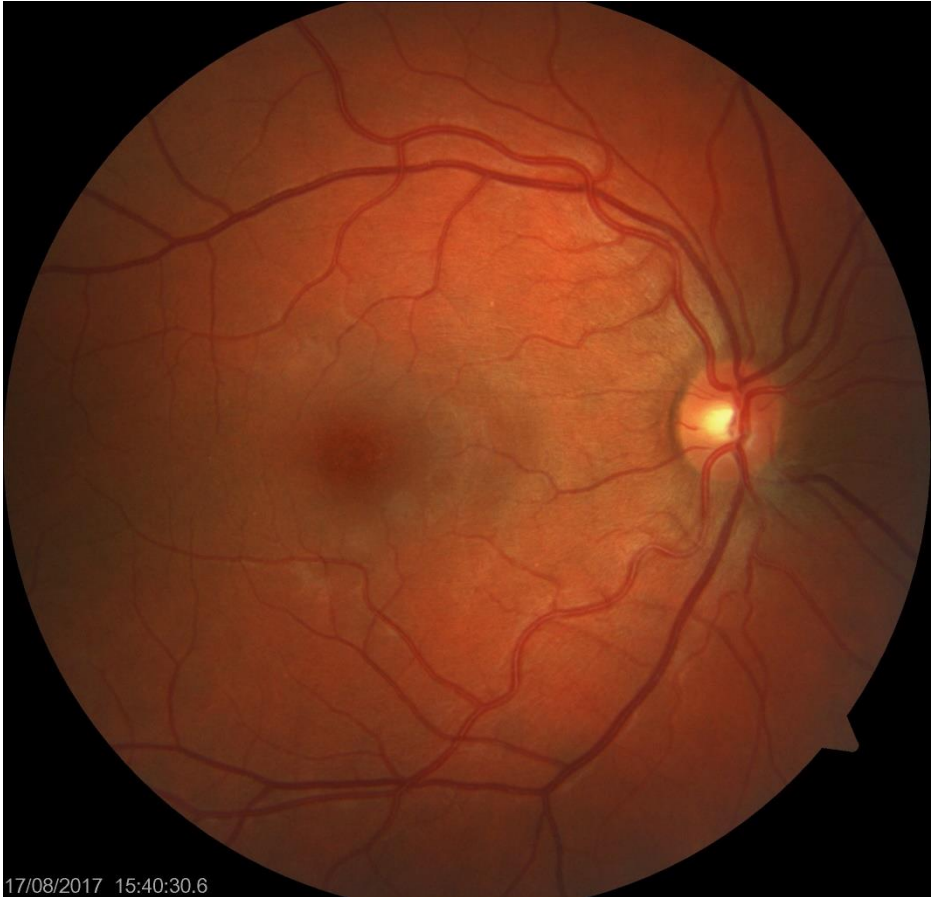


Diagnosis:



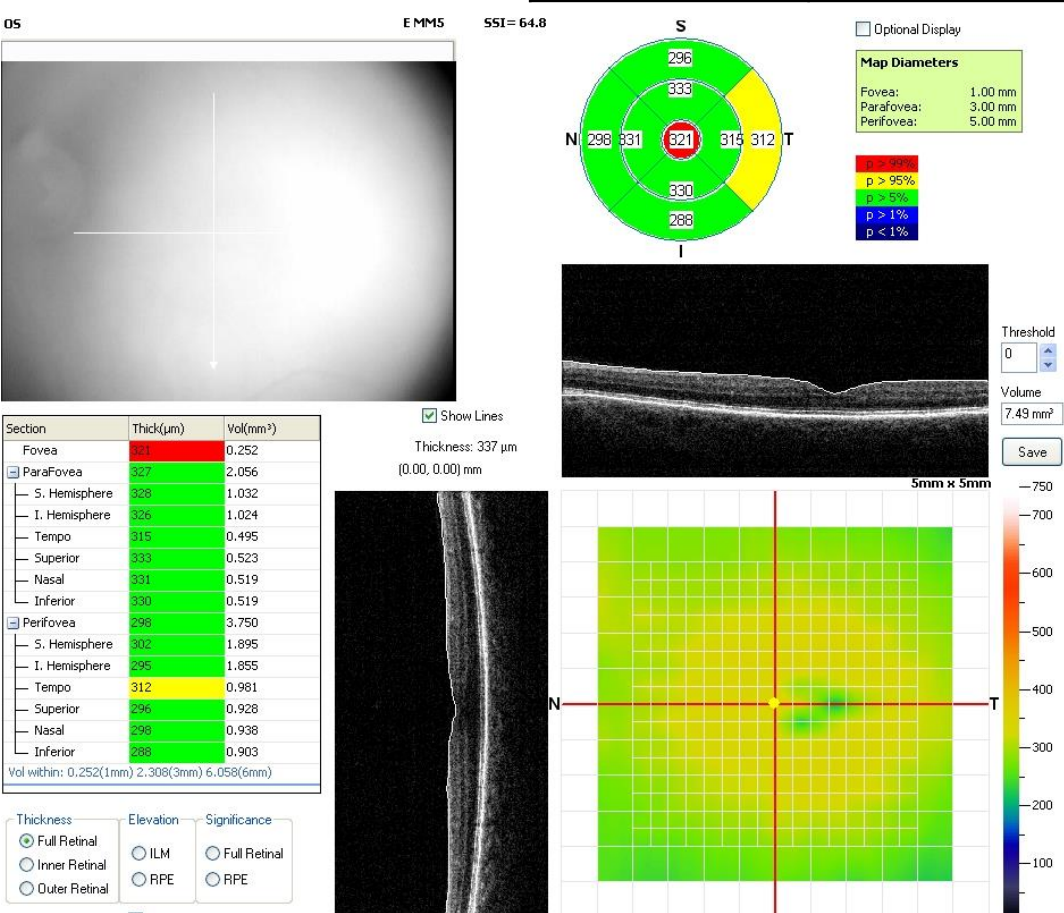
Jan, 46 jaar

Jaar DM	2 jaar
HbA1C	6,3%
Bloeddruk	geen
Cholesterol	geen



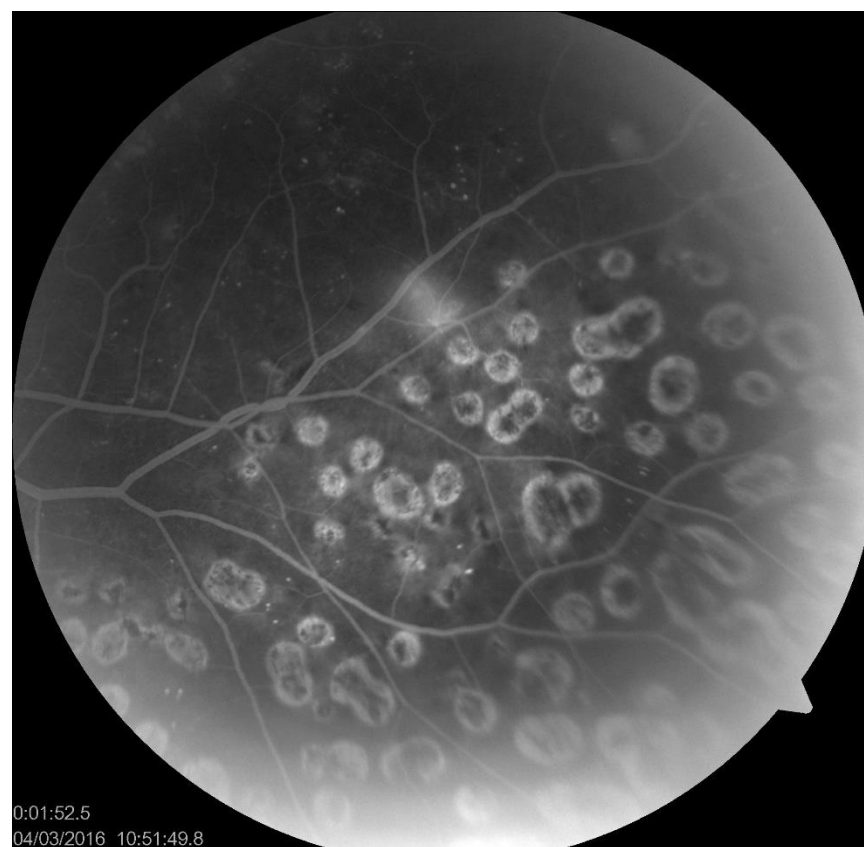
Vera, 47 jaar

Jaar DM	2 jaar
HbA1C	8,4%
Bloeddruk	15/9
Cholesterol	medicatie



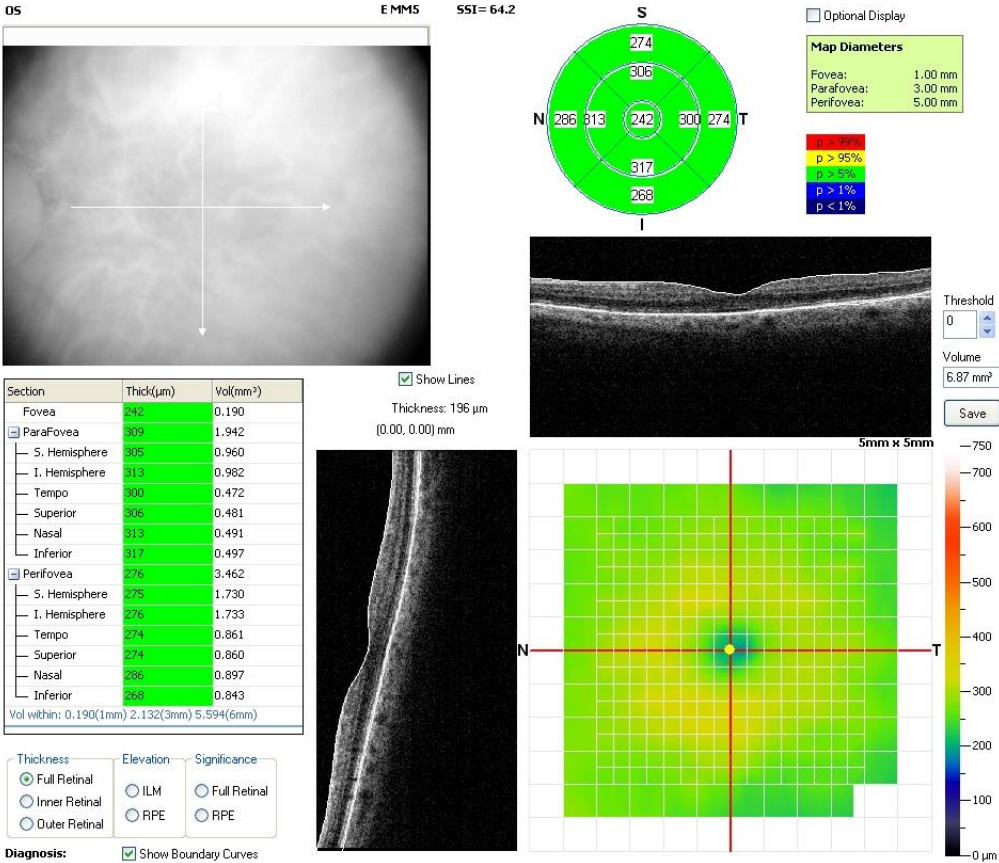
Johan, 60 jaar

Jaar DM	20 jaar
HbA1C	7,8%
Bloeddruk	medicatie
Cholesterol	medicatie



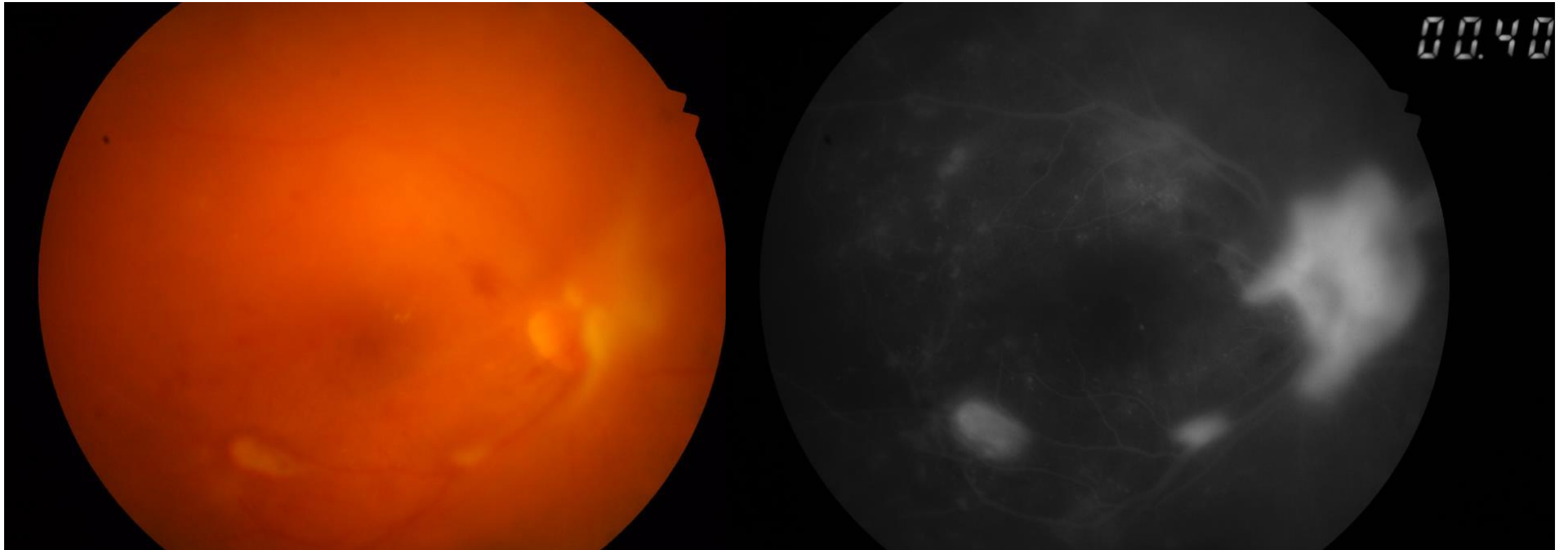
Janine, 35 jaar, zwanger

Jaar DM	5 jaar
HbA1C	6,5%
Bloeddruk	geen
Cholesterol	geen



Marina, 57 jaar

Jaar DM	17 jaar
HbA1C	8,5%
Bloeddruk	medicatie
Cholesterol	neemt geen



RETINA RISK

ABOUTRISK CALCULATOREDUCATIONAL MATERIALPRICING



✔Data

✔Step 1
Patient information

✔Step 2
Type of information

✔Results
Type of information

Clinical data: Patient information

Please enter your clinical data. Leave boxes empty if values are unknown.

DURATION OF DIABETES

01020304010 years

Note that you fill in either HbA1c or Average whole blood glucose, not both.

✔HBA1C VALUES (PERCENTAGE)

0369127%

HBA1C VALUES (POINTS)

01000

✔AVERAGE WHOLE BLOOD GLUCOSE

SYSTOLIC BLOOD PRESSURE

0180130 mmHg

DIASTOLIC BLOOD PRESSURE

07212080 mmHg

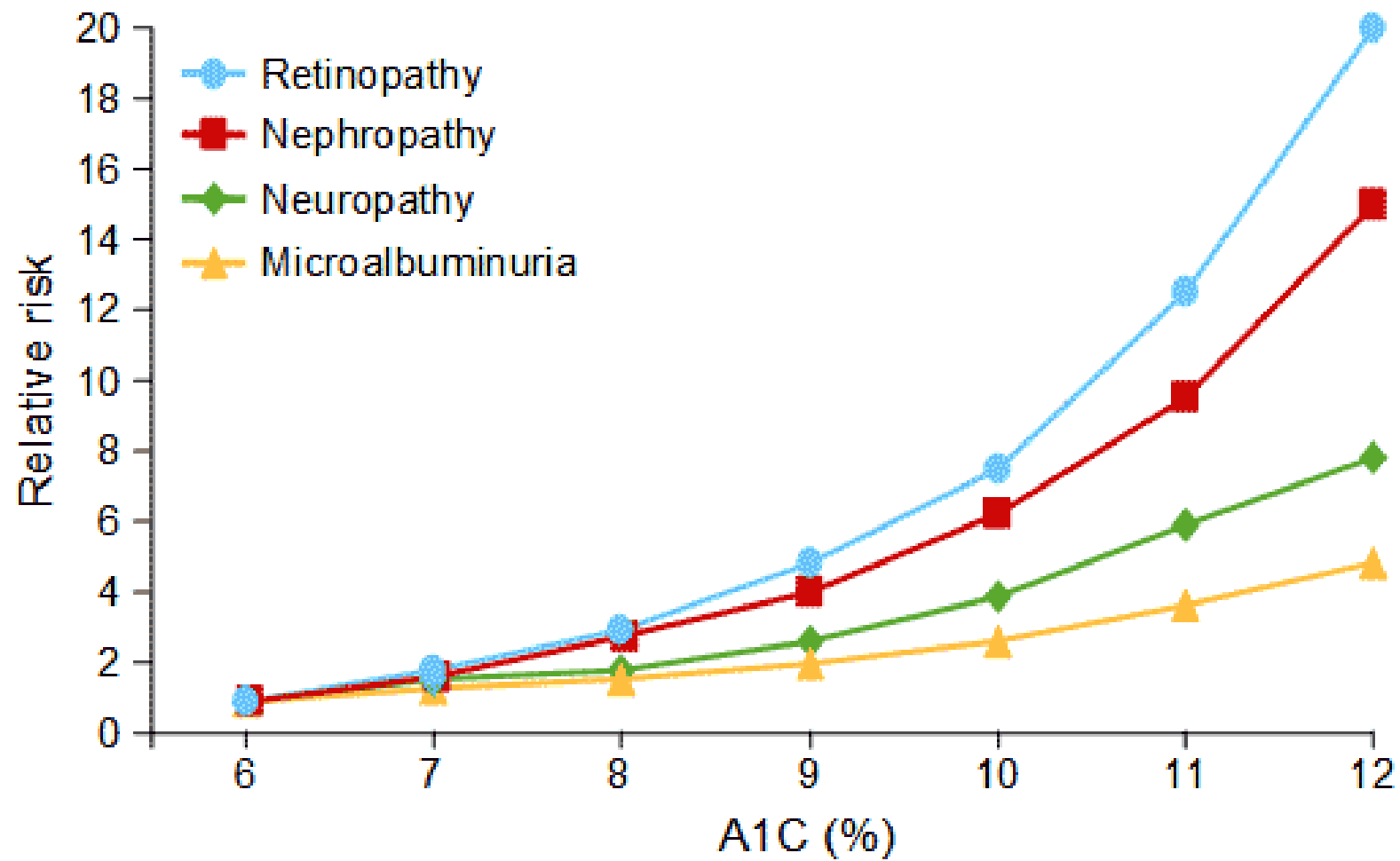
- Based on your clinical data, we evaluate that you are in **medium risk** of developing sight-threatening retinopathy.
- Your calculated risk of developing sight-threatening retinopathy is **2.3%** every year.
- Your HbA1c is too high and impacts your progression the most. Keep your HbA1c levels less than 6.5% in consultation with your diabetes physician.



Oogkliniek Jessaziekenhuis 2017

- 211 patiënten
- 28 type 1 DM en 171 type 2 DM (12 geen gegevens)

HbA1C %	(n)	Geen drp	Milde	Matige	Ernstige	Proliferatieve
<7	55	31	11	3	0	(1)
7-8	44	20	20	4	0	(1)
>8	45	14	17	10	2	2



Oogkliniek Jessaziekenhuis 2017

HbA1C %	(n)	Maculopathie
<7	55	7
7-8	44	6
>8	45	14

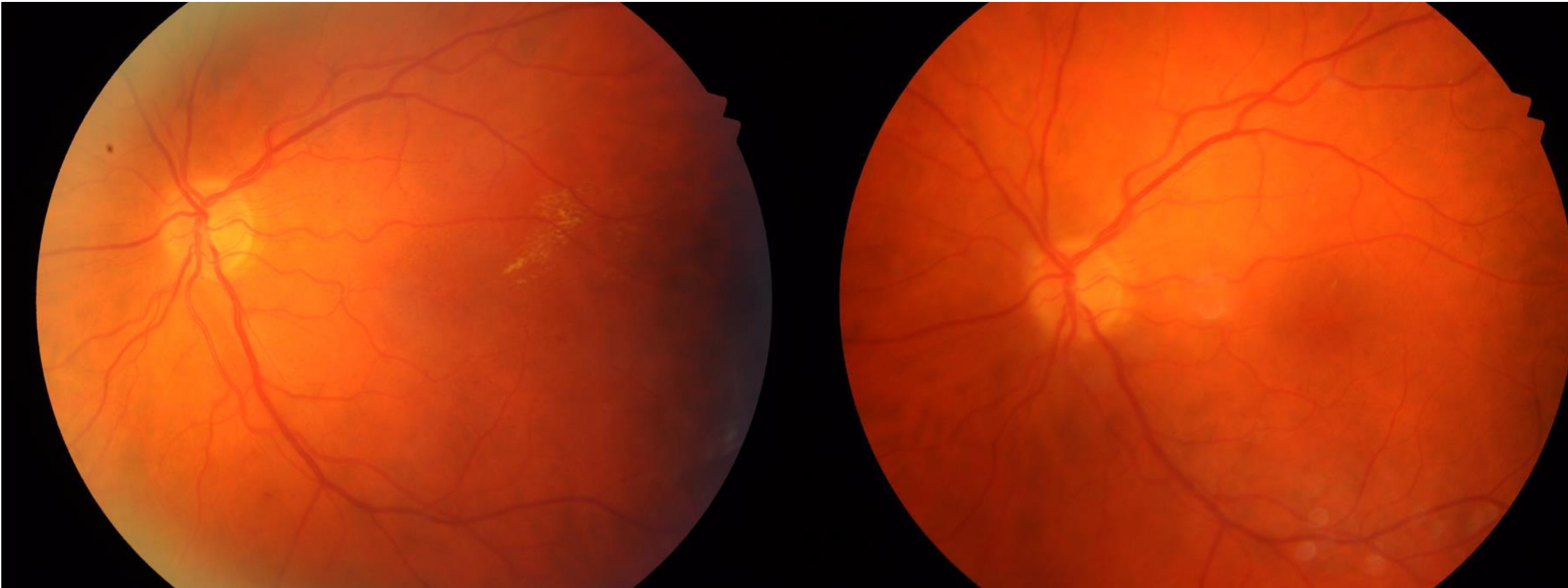
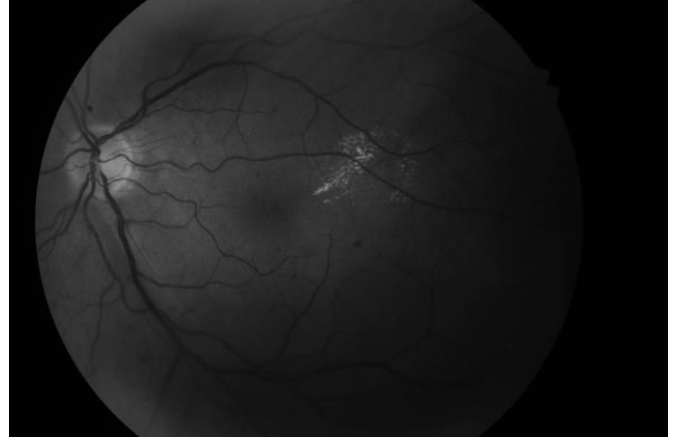
Oogkliniek Jessaziekenhuis 2017

HbA1C %	(n)	Laser	Intravitreale injectie
<7	55	5	3
7-8	44	11	4
>8	45	14	10

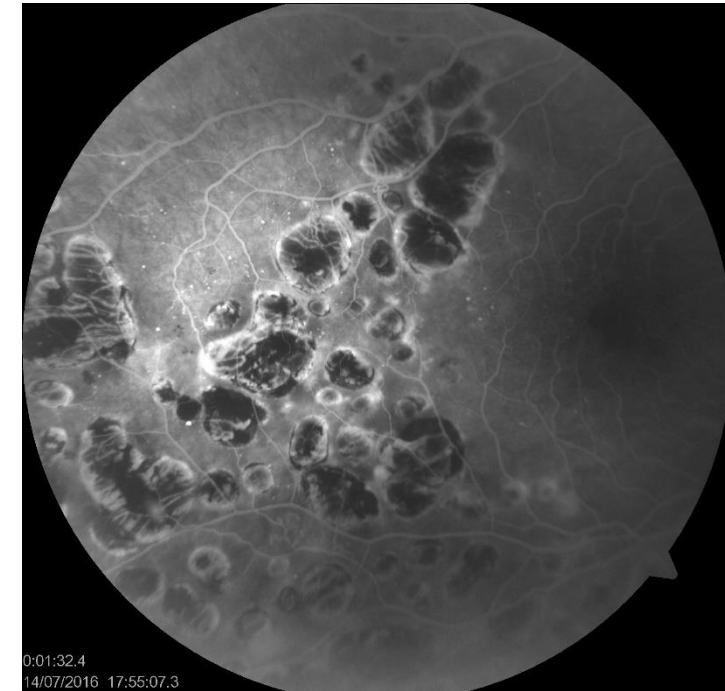
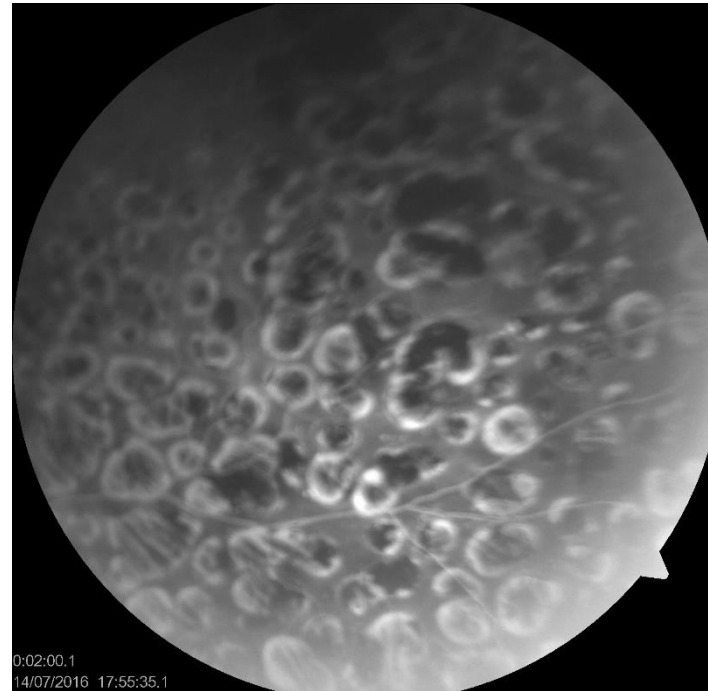
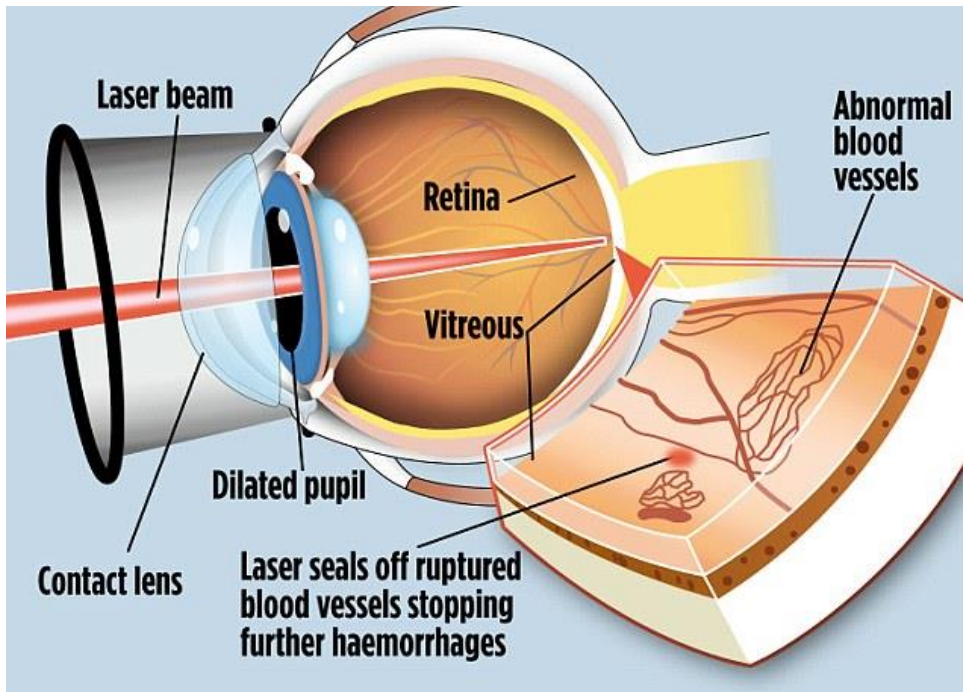
Oogkliniek Jessaziekenhuis 2017

Aantal jaar	(n)	Geen drp	Milde	Matige	Ernstige	Proliferatieve
<10	57	38	9	7	1	2
10-20	59	29	21	4	1	4
>20	37	12	15	8	0	1

Behandeling: Afwachten



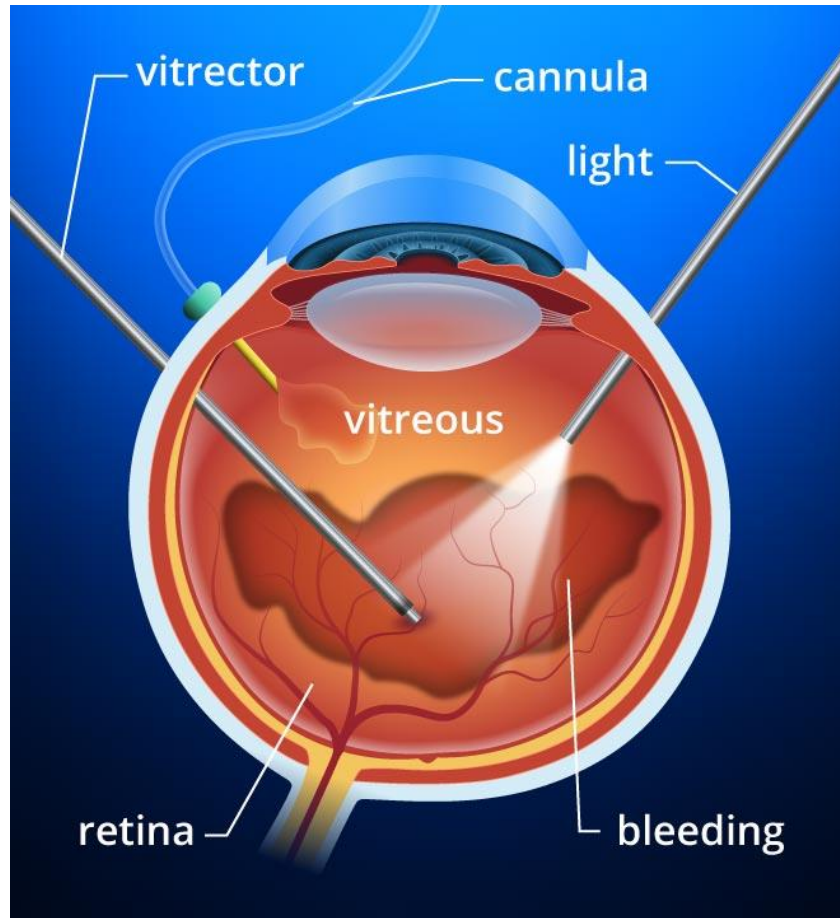
Behandeling: laser



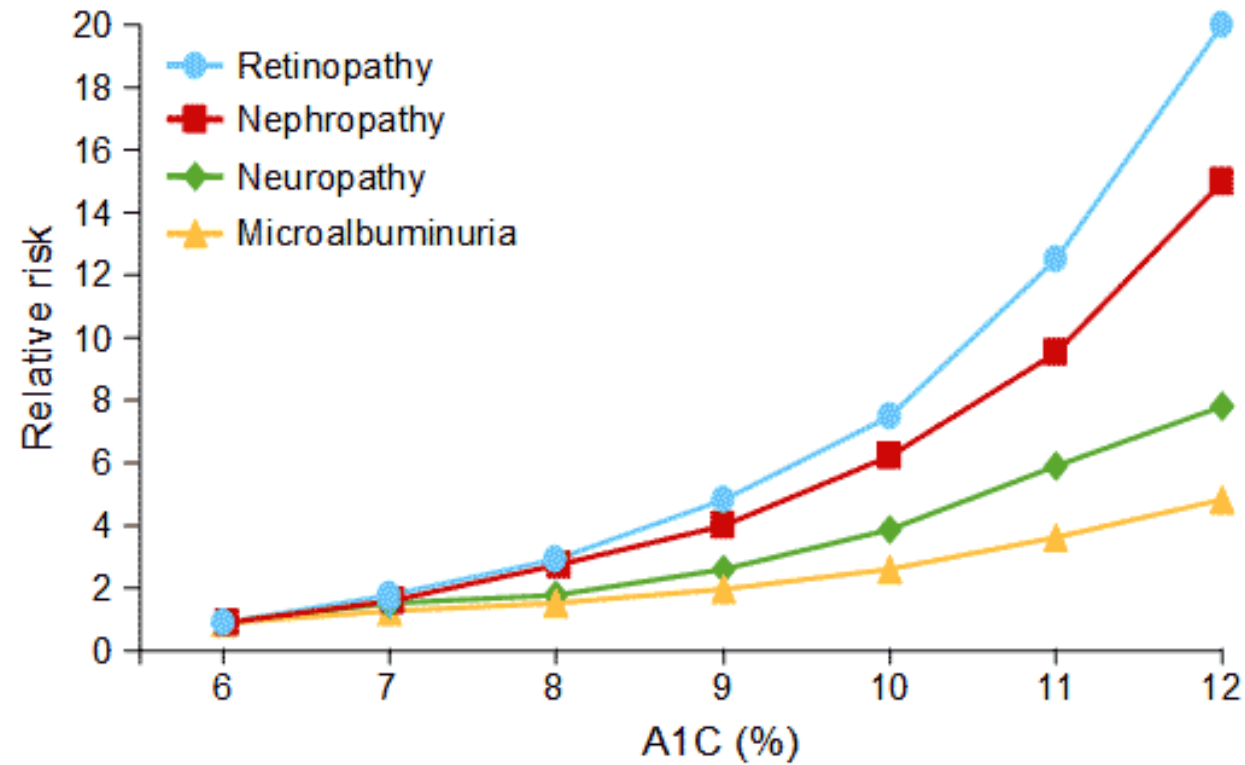
Behandeling: intravitreale injecties

- Intravitreale steroïden: Triamcinolone/Dexametasone (Ozurdex)/kenacort/volon
 - Oogdrukstijging
 - Cataract
- Anti-angiogenetische medicatie: bevacizumab (Avastin), ranibizumab (Lucentis), aflibercept (Eylea). ? Iedere 2 maand ? Combinatie met laser

Behandeling: vitrectomie



Preventie = Glycemie controle!



Risicofactoren Diabetische retinopathie

- Type 1 DM heeft 2,5 meer kans dan Type 2 DM (onafhankelijk van duur DM)
- Duur van de DM
- HbA1C ($> 6,5\%$) [let op: niet te snel corrigeren]
- Hypertensie (3 keer meer kans $> 140\text{mmHg}$)
- Cholesterol (R/ fibraten)
- Zwangerschap
- Puberteit
- Meer bij afro-amerikanen
- Slaapapnee (CPAP)

Preventie

Aanbeveling van de American Academy of Ophthalmology & American Diabetes Association:

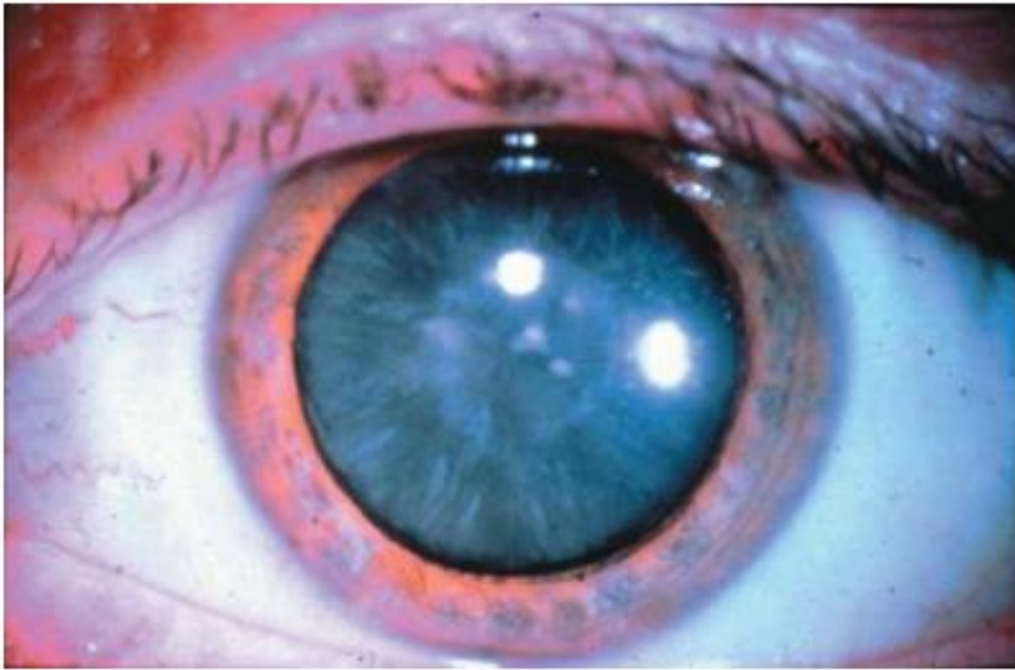
Jaarlijks gedilateerde fundusonderzoek door een oogarts

Bij kinderen vanaf 9 jaar en 5 jaar DM

Bij zwangeren met gekende DM: iedere 3 maand. (maandelijks bij DRP)

Cataractchirurgie en maculopathie

- Nepafenac (Nevanac) druppels



Dank voor uw aandacht

Dank voor de uitnodiging van de diabetesliga en de organisatoren

- info@oogartsmissotten.be